

CONFLICT DELETION AND ORDERED ENDPOINTS IN RANDOM INCOMING-COPY SYMMETRIZATION

ANONYMOUS

ABSTRACT. On a loopless labelled directed graph, choose a uniform vertex and replace its outgoing star by a copy of its incoming star. We determine the resulting absorbing chain at every time. The unordered pairs carrying unequal opposite arcs form a conflict graph, and one update deletes precisely the conflict star at the chosen vertex. Hence the length- t absorption probability is an exact sum over independent vertex sets of the initial conflict graph. The endpoint law is finer: the local idempotents do not commute, and each resolved conflict is oriented by which endpoint appears first in the history. We turn this first-occurrence order into a complete source–target kernel using Stirling numbers. Finally, for every target we count both labelled predecessor/action pairs and distinct predecessor states; the two counts differ because all admissible predecessor families meet at the target. Exhaustion through four vertices supplies 47,033 author-side regression assertions, not proof or novelty evidence. External ownership remains unresolved, so the manuscript status is `HOLD_EXTERNAL`.

1. THE CHAIN AND ITS SUBTRACTION BOUNDARY

Fix $n \geq 1$. Let $\mathcal{D}_n$ be the loopless directed graphs on $[n] = \{1, \dots, n\}$, represented by independent bits $A_{uv} \in \{0, 1\}$ for $u \neq v$. Both opposite arcs may be present. For $v \in [n]$, define $C_v : \mathcal{D}_n \rightarrow \mathcal{D}_n$ by

$$(1) \quad (C_v A)_{vu} = A_{uv} \quad (u \neq v), \quad (C_v A)_{ij} = A_{ij} \quad (i \neq v).$$

Thus v copies its incoming star to its outgoing star. The Markov chain chooses v uniformly and has transition operator

$$(2) \quad P = \frac{1}{n} \sum_{v=1}^n C_v.$$

Reciprocal arcs are standard statistics of directed-network models. Yin–Zhu study their equilibrium ensembles [3], while Cirkovic–Wang–Resnick add reciprocal edges inside a growing preferential attachment process [2]. Those models do not use the fixed-carrier incoming-star copy (1). General Markov chains generated by finite semigroups are also established background [1]. We assign reciprocity as a statistic, generic finite semigroup walks, coupon support, Stirling enumeration, and independence polynomials zero contribution credit. The retained object is the conjunction of conflict deletion, the independent-set absorption law, the ordered source–target kernel, and the dual inverse census for the literal chain. A later literal or equivalent owner triggers withdrawal; a bounded search non-hit is not a novelty certificate.

The closest internal comparison is P179. Its singleton-isolation maps are commuting idempotents, so a history acts only through its support. Here $C_u C_v \neq C_v C_u$ whenever $\{u, v\}$ is a conflict: the endpoint appearing first chooses the surviving common bit. P145 toggles undirected vertex cuts, P159 prunes through degree parity, and P177 is an additive Cayley walk. None has the monotone conflict deletion plus ordered endpoint kernel below. These differences are an internal collision subtraction, not an external priority claim.

2. CONFLICT DELETION AND RECURRENCE

Associate to $A \in \mathcal{D}_n$ the simple *conflict graph*

$$(3) \quad H(A) = ([n], \{\{u, v\} : A_{uv} \neq A_{vu}\}).$$

An isolated vertex is understood relative to this graph, including when $n = 1$.

Lemma 2.1 (exact conflict deletion). *For every $A \in \mathcal{D}_n$ and $v \in [n]$,*

$$(4) \quad E(H(C_v A)) = \{e \in E(H(A)) : v \notin e\}.$$

Consequently C_v is idempotent, but the family $(C_v)_{v \in [n]}$ need not commute.

Key words and phrases. finite Markov chain, directed graph, reciprocity, absorption, independent set, fibre.

Proof. For each $u \neq v$, equation (1) makes the two arcs on $\{u, v\}$ equal to the old bit A_{uv} . A pair not incident with v is unchanged. This proves (4); applying the same copy again has no effect. If $A_{uv} \neq A_{vu}$, applying C_u first makes both bits equal to A_{vu} , whereas applying C_v first makes both equal to A_{uv} . Thus the two possible first moves give different resolved pairs. $\square$

Corollary 2.2 (recurrent states). *The recurrent states are exactly the $2^{\binom{n}{2}}$ symmetric digraphs, and each is fixed.*

Proof. A symmetric digraph has no conflict and every update fixes it. If a state has a conflict edge, selecting either endpoint deletes that edge, and no update can recreate any deleted conflict. A path of positive probability therefore leaves its conflict level and cannot return. Such a state is transient in the finite chain. $\square$

3. INDEPENDENT-SET ABSORPTION LAW

Let $W = (w_1, \dots, w_t) \in [n]^t$ be a vertex history, with support $\text{supp}(W)$ and missing set $M(W) = [n] \setminus \text{supp}(W)$. Write τ for the first epoch at which the state is symmetric. Let $\left\{ \begin{smallmatrix} t \\ r \end{smallmatrix} \right\}$ be a Stirling number of the second kind, with $\left\{ \begin{smallmatrix} 0 \\ 0 \end{smallmatrix} \right\} = 1$.

Lemma 3.1 (history residue). *For every initial state A and history W ,*

$$(5) \quad E(H(C_{w_t} \cdots C_{w_1} A)) = E(H(A)[M(W)]).$$

Proof. Apply Lemma 2.1 at each epoch. Selecting a vertex removes all incident conflicts, and repeating a selected vertex removes nothing further. The edges left after the complete history are precisely those whose two endpoints were never selected, which are the edges induced by $M(W)$. $\square$

Theorem 3.2 (all-time absorption CDF). *For $A \in \mathcal{D}_n$ and every integer $t \geq 0$,*

$$(6) \quad \Pr_A(\tau \leq t) = \frac{1}{n^t} \sum_{M \in \mathcal{I}(H(A))} (n - |M|)! \left\{ \begin{smallmatrix} t \\ n - |M| \end{smallmatrix} \right\},$$

where $\mathcal{I}(H)$ is the family of independent vertex sets of H . The formula includes $t = 0$ and the already symmetric case.

Proof. By Lemma 3.1, absorption by time t is equivalent to $H(A)[M(W)]$ having no edge, or to $M(W)$ being independent. For a fixed missing set M , the histories with exact support $[n] \setminus M$ are the surjections from $[t]$ onto $n - |M|$ prescribed labels. Their number is $(n - |M|)! \left\{ \begin{smallmatrix} t \\ n - |M| \end{smallmatrix} \right\}$. Different missing sets define disjoint history classes. Summing their counts and dividing by the n^t equiprobable words proves (6). At $t = 0$, only $M = [n]$ can contribute, and it does so exactly when the initial conflict graph is empty. $\square$

Thus the temporal law remembers the entire independent-set inventory of the initial conflict graph, not only its number of edges or nonisolated vertices.

4. FIRST-OCCURRENCE ENDPOINT KERNEL

For $S \subseteq [n]$, let $\pi = (\pi_1, \dots, \pi_r)$ be a total order of S . Give a vertex outside S rank $+\infty$. Define $E_{S,\pi}(A)$ pairwise as follows. A nonconflict remains unchanged. A conflict $\{u, v\}$ disjoint from S also remains unchanged. Otherwise let e be its finite-rank endpoint of smaller rank and let o be the other endpoint; replace both opposite arcs by the old incoming bit A_{oe} .

Theorem 4.1 (every source–target multiplicity). *Let $N_t(A, B)$ be the number of histories $W \in [n]^t$ sending A to B . Then*

$$(7) \quad N_t(A, B) = \sum_{\substack{S \subseteq [n], \pi \in \mathfrak{S}(S) \\ E_{S,\pi}(A) = B}} \left\{ \begin{smallmatrix} t \\ |S| \end{smallmatrix} \right\}.$$

Hence $P^t(A, B) = N_t(A, B)/n^t$ for every labelled source, target, and time.

Proof. Fix a history. Lemma 2.1 shows that the first selected endpoint of a conflict makes its two arcs equal; later updates preserve their equality. Equation (1) shows that the common value is the old arc entering that first endpoint. The history endpoint is therefore exactly $E_{S,\pi}(A)$ for its support S and first-occurrence order π .

It remains to count histories with a prescribed pair (S, π) . Partition the time positions according to which of the $r = |S|$ labels occurs there. Ordering these nonempty blocks by their least positions and labelling them $\pi_1, \dots, \pi_r$ is a bijection from set partitions of $[t]$ into r blocks to histories with that

first-occurrence order. There are $\{^t_r\}$ such histories. The support/order classes are disjoint, so summing those producing B proves (7). $\square$

5. COMPLETE ONE-STEP INVERSE CENSUS

For a target B , let $k(B)$ be the number of isolated vertices of $H(B)$.

Theorem 5.1 (labelled actions versus distinct sources). *The number of labelled pairs $(A, v) \in \mathcal{D}_n \times [n]$ satisfying $C_v A = B$ is*

$$(8) \quad k(B)2^{n-1}.$$

The number of distinct source states is

$$(9) \quad \#\{A : \exists v, C_v A = B\} = \begin{cases} 0, & k(B) = 0, \\ 1 + k(B)(2^{n-1} - 1), & k(B) > 0. \end{cases}$$

Proof. If $C_v A = B$, all pairs incident with v are symmetric in B , so v must be isolated in $H(B)$. Conversely, fix such a vertex. Every arc not leaving v is forced to equal its target bit, while the $n - 1$ arcs leaving v are overwritten and may be chosen freely. This gives 2^{n-1} sources for each of the $k(B)$ labelled actions, proving (8).

Each admissible-vertex source family contains B . Suppose a source belongs to the families for two distinct vertices v and w . Membership in the w -family forces every arc leaving v except possibly $v \rightarrow w$ to its target value; the v -family gives the analogous statement for arcs leaving w . On the remaining pair, the v -family fixes $w \rightarrow v$ and the w -family fixes $v \rightarrow w$. Thus the source is B . The families have no other intersections, so their union contains one shared state plus $2^{n-1} - 1$ new states per isolated vertex. If there is no isolated vertex, there is no admissible action. This proves (9). $\square$

For $n = 1$, the unique empty digraph is fixed, has one labelled predecessor pair and one distinct source; all formulas include this boundary.

6. EXACT CONTROL, LIMITATIONS, AND CONCLUSION

The accompanying standard-library program exhausts all loopless digraphs through $n = 4$, all conflict graphs through $n = 4$, and all histories of length at most n . Table 1 records selected outputs. These checks apply counterexample pressure to the formulas; the proofs above support the unbounded statements.

n	states	recurrent	max. distinct fibre	absorbed words from K_n at $t = n$
1	1	1	1	1
2	4	2	3	4
3	64	8	10	24
4	4096	64	29	168

TABLE 1. Author-side exhaustive controls. The K_n row starts from one asymmetric arc pair on every edge.

The analysis assumes labelled vertices, binary loopless arcs, and uniform single-vertex updates. It does not classify endpoint multiplicities by graph isomorphism, give asymptotics for arbitrary graph families, or cover weighted, looped, or nonuniform variants. External ownership has received only a bounded screen. These limits keep the status at `HOLD_EXTERNAL`.

DECLARATIONS

Data availability. No external data were used. The finite regression program and its canonical output accompany the manuscript.

Ethics. The work studies abstract finite graphs and raises no human-subject, personal-data, or deployment issue.

CRedit. Anonymous author(s) performed conceptualization, formal analysis, software, validation, original drafting, and review and editing.

Competing interests. No competing interests are declared.

Funding. No external funding is declared for this manuscript.

AI-use statement. Generative AI tools assisted with ideation, drafting, and code scaffolding. The author-side verifier checks finite cases; responsibility for every definition, proof, citation, and disclosure remains with the human author(s).

Release status. External circulation, posting, and submission are not authorized while `HOLD_EXTERNAL` remains active.

REFERENCES

- [1] Kenneth S. Brown. Semigroups, rings, and Markov chains. *Journal of Theoretical Probability*, 13(3): 871–938, 2000. doi: 10.1023/A:1007822931408.
- [2] Daniel Cirkovic, Tiandong Wang, and Sidney I. Resnick. Preferential attachment with reciprocity: Properties and estimation. *Journal of Complex Networks*, 11(5):cnad031, 2023. doi: 10.1093/comnet/cnad031.
- [3] Mei Yin and Lingjiong Zhu. Reciprocity in directed networks. *Physica A: Statistical Mechanics and its Applications*, 447:71–84, 2016. doi: 10.1016/j.physa.2015.12.008.